

The floating-function affine isoperimetric conjecture is false

Counterexample packet for arXiv:1711.11088

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Counterexample in every dimension. For each $n \geq 1$, the smooth, even, strongly convex functions

$$\psi_\varepsilon(x) = \frac{|x|^2}{2} + \varepsilon x_1^4$$

violate the conjectured affine isoperimetric inequality for all sufficiently small $\varepsilon > 0$. Indeed, the normalized ratio equals one at $\varepsilon = 0$ and has derivative $6/(n+2) > 0$ there. Thus the Gaussian is not even a local maximizer for the affine surface area introduced in the source.

The source conjecture

For a convex function $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$ with $0 < \int e^{-\psi} < \infty$, Li, Schütt, and Werner define the affine surface area of $f = e^{-\psi}$ by

$$\text{as}(f) = \int_{\mathbb{R}^n} (\det \nabla^2 \psi(x))^{1/(n+2)} e^{-\psi(x)} dx.$$

After proving the inequality for 2-homogeneous ψ , they conjecture that it holds for every convex ψ [1]:

$$\text{as}(f) \leq (2\pi)^{n/(n+2)} \left(\int_{\mathbb{R}^n} e^{-\psi(x)} dx \right)^{n/(n+2)}. \quad (1)$$

$$\int_{\mathbb{R}^n} e^{-(\frac{n}{n+2}\psi(x) + \frac{1}{n+2}\langle x, \nabla \psi(x) \rangle)} (\det \nabla^2 \psi(x))^{\frac{1}{n+2}} dx. \quad (16)$$

Both definitions coincide for 2-homogeneous functions ψ . From an affine isoperimetric inequality proved in [18] for the expression (16) we can deduce the following corollary.

Corollary 1. *Let $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$ be a 2-homogeneous, convex function such that $0 < \int_{\mathbb{R}^n} e^{-\psi(x)} dx < \infty$. Then*

$$\text{as}(f) \leq (2\pi)^{\frac{n}{n+2}} \left(\int_{\mathbb{R}^n} e^{-\psi(x)} dx \right)^{\frac{n}{n+2}},$$

with equality if and only if there are $a \in \mathbb{R}$ and a positive definite matrix A such that for all $x \in \mathbb{R}^n$

$$\psi(x) = \langle Ax, x \rangle + a.$$

We conjecture that this inequality holds for general, convex functions.

We include the argument for (15) for completeness.

We integrate in polar coordinates with respect to the normalized cone measure σ_K of K .

We give a variational counterexample that retains the most natural normalizations: ψ is even, C^∞ , strongly convex, and has minimum zero at the origin.

A quartic perturbation of the Gaussian

Fix $n \geq 1$. For $\varepsilon \geq 0$, let

$$\psi_\varepsilon(x) = \frac{|x|^2}{2} + \varepsilon x_1^4, \quad Z(\varepsilon) = \int_{\mathbb{R}^n} e^{-\psi_\varepsilon(x)} dx.$$

Since

$$\nabla^2 \psi_\varepsilon(x) = \text{diag}(1 + 12\varepsilon x_1^2, 1, \dots, 1),$$

the corresponding affine surface area is

$$A(\varepsilon) = \int_{\mathbb{R}^n} (1 + 12\varepsilon x_1^2)^{1/(n+2)} e^{-|x|^2/2 - \varepsilon x_1^4} dx.$$

Normalize the two sides of (1) by setting

$$R(\varepsilon) = \frac{A(\varepsilon)}{(2\pi)^{n/(n+2)} Z(\varepsilon)^{n/(n+2)}}.$$

The conjecture asserts $R(\varepsilon) \leq 1$. At $\varepsilon = 0$,

$$A(0) = Z(0) = (2\pi)^{n/2}, \quad R(0) = 1.$$

Both integrals are right-differentiable at zero. This follows directly from dominated convergence: on any bounded interval of nonnegative ε , the differentiated integrands are bounded by a fixed polynomial times $e^{-|x|^2/2}$. If G is a standard Gaussian vector, then $\mathbb{E}G_1^2 = 1$ and $\mathbb{E}G_1^4 = 3$, and hence

$$\begin{aligned} \frac{Z'(0)}{Z(0)} &= -\mathbb{E}G_1^4 = -3, \\ \frac{A'(0)}{A(0)} &= \mathbb{E} \left(\frac{12}{n+2} G_1^2 - G_1^4 \right) = \frac{12}{n+2} - 3. \end{aligned}$$

It follows that

$$\begin{aligned} \left. \frac{d}{d\varepsilon} \log R(\varepsilon) \right|_{0+} &= \frac{A'(0)}{A(0)} - \frac{n}{n+2} \frac{Z'(0)}{Z(0)} \\ &= \frac{12}{n+2} - 3 + \frac{3n}{n+2} = \frac{6}{n+2} > 0. \end{aligned}$$

Therefore

$$R(\varepsilon) = 1 + \frac{6}{n+2} \varepsilon + o(\varepsilon) \quad (\varepsilon \downarrow 0). \quad (2)$$

Theorem 1. *For every $n \geq 1$, there exists $\varepsilon_n > 0$ such that every $0 < \varepsilon < \varepsilon_n$ gives a smooth, even, strongly convex function $\psi_\varepsilon(x) = |x|^2/2 + \varepsilon x_1^4$ for which*

$$\text{as}(e^{-\psi_\varepsilon}) > (2\pi)^{n/(n+2)} \left(\int_{\mathbb{R}^n} e^{-\psi_\varepsilon(x)} dx \right)^{n/(n+2)}.$$

Consequently, the conjecture following Corollary 1 of the source is false in every dimension.

Indeed, the conclusion follows immediately from (2). Notice that $\psi_\varepsilon(0) = \min \psi_\varepsilon = 0$, so the failure cannot be removed merely by fixing the additive constant in the potential.

A still simpler literal obstruction

The formulation for general convex functions is also incompatible with additive constants. Let

$$\psi_a(x) = \frac{|x|^2}{2} + a.$$

Its normalized ratio is

$$R_a = e^{-2a/(n+2)}.$$

Thus every $a < 0$ already violates (1). In the source's proved corollary, 2-homogeneity forces $a = 0$ despite the displayed equality description allowing an additive constant. The quartic family above is the substantive obstruction: it remains after imposing minimum zero and changes the shape rather than the height of the Gaussian.

Adversarial checks and scope

1. The Hessian is positive definite everywhere for $\varepsilon \geq 0$, so no generalized-Hessian or boundary convention is involved.
2. The perturbation is integrable and all differentiated integrands admit Gaussian domination. The proof uses an exact first variation, not a numerical approximation.
3. Equation (2) is strict in the forbidden direction, so continuity supplies actual positive parameters, not only a formal infinitesimal obstruction.
4. The example is even and centered. It therefore also survives the common barycenter-at-zero hypothesis used in related functional affine inequalities.
5. The counterexample concerns the source's new affine surface area, whose weight is $e^{-\psi}$. It does not contradict the different functional affine isoperimetric inequality for expression (16) in the source.

Bounded novelty check

On 11 August 2026, the exact title and conjecture wording, the Hessian-determinant functional, Gaussian extremizers, floating functions, and affine isoperimetric inequalities for log-concave functions were checked in the run indexes and by bounded web/arXiv search. The searches found the source, its later Ulam and weighted floating-function developments, and related but different functional affine surface areas; no answer or counterexample to this exact conjecture was located. This is evidence for novelty, not a definitive guarantee.

References

- [1] B. Li, C. Schütt, and E. M. Werner, *Floating functions*, Israel J. Math. **231** (2019), 181–210; arXiv:1711.11088.